Period 2
Member: Nusaibah Karim
Group Name: Nusaibah Karim Final Project
Project Title: Geometry Dash Recreation

Description

My project is based on a popular game called Geometry Dash. It is a single-player game, where the player is a rotating square that moves across the screen. The objective of the original game is to overcome obstacles, like jump sequences, pits, and spikes, to reach the final checkpoint. I will replicate this game with the following logic.

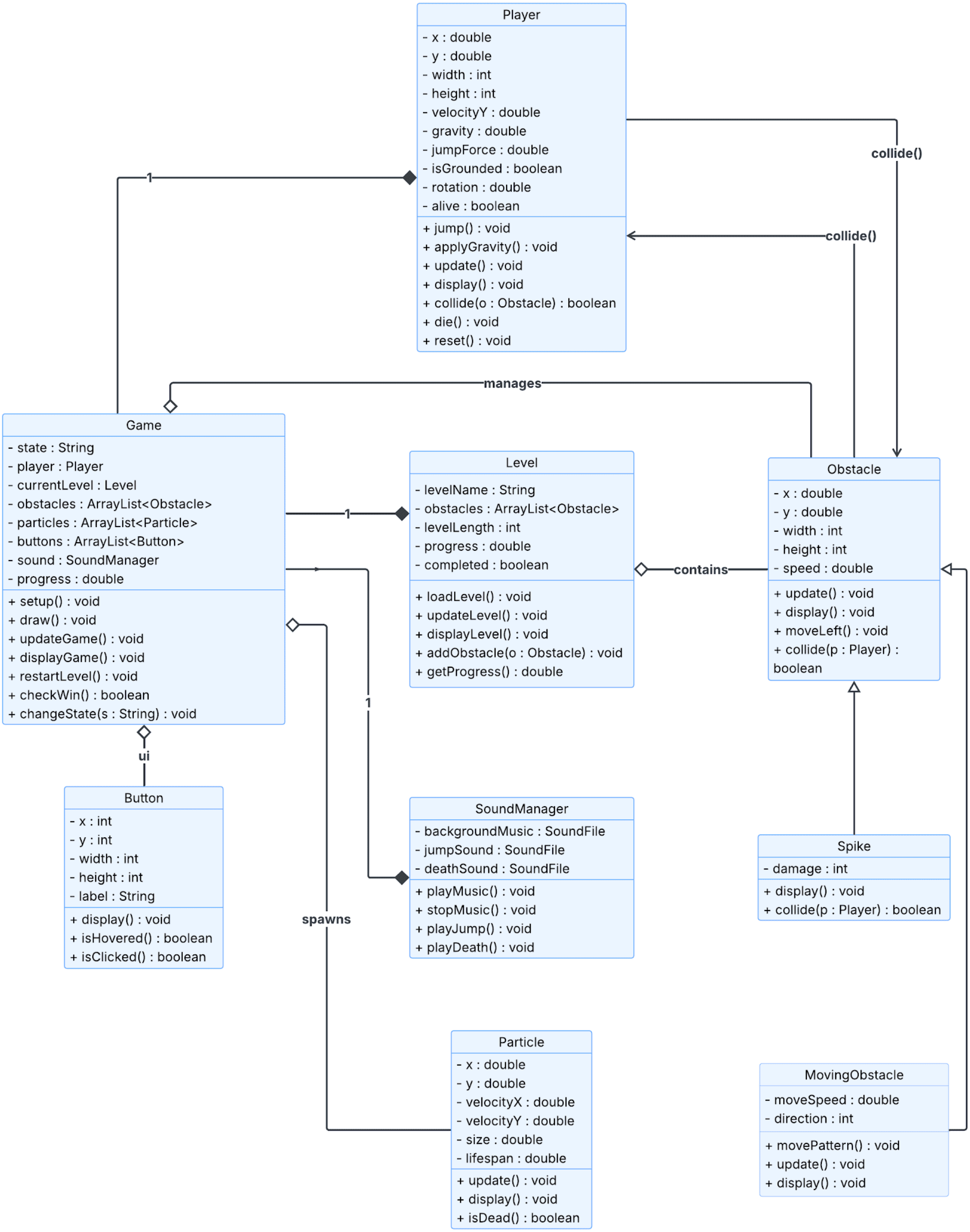
At the start of the game, there will be a homescreen, where the player has to click a button to start the game. The player will need to press the up arrow on their keyboard to jump over various obstacles. The game includes multiple types of obstacles, all of which inherit from an Obstacle class. As the game progresses, the obstacles will increase in variety to add difficulty to the level. Some of these obstacles include moving platforms and spikes. There will be collision detection with the square, so the player dies if they touch the obstacle. There will be a gravity force in the game, affecting the height of the player's jumps.

Other additions I would like to make to the game are special screens to indicate winning or losing. The user will be able to track their progress with the progress bar at the top of the screen.

I plan on using the Processing Sound Library to handle all audio features in the game. There will be background music during each level, unique sound effects triggered when the player jumps, and a death sound when the player collides with an obstacle.

How does it work?

The game starts when the user clicks the run button on the screen. The up arrow is used to jump. In the game itself, the player will move forward automatically with a side-scrolling effect. The user will need to avoid obstacles and spikes by clicking the up arrow to jump. The goal is to reach 100% progress to win the level. If any obstacles are touched, then the user dies, and the player returns to the homescreen.


Player
- x : double
- y : double
- width : int
- height : int
- velocityY : double
- gravity : double
- jumpForce : double
- isGrounded : boolean
- rotation : double
- alive : boolean
+ jump() : void
+ applyGravity() : void
+ update() : void
+ display() : void
+ collide(o : Obstacle) : boolean
+ die() : void
+ reset() : void
collide()
collide()
1
manages
Game
- state : String
- player : Player
- currentLevel : Level
- obstacles : ArrayList<Obstacle>
- particles : ArrayList<Particle>
- buttons : ArrayList<Button>
- sound : SoundManager
- progress : double
+ setup() : void
+ draw() : void
+ updateGame() : void
+ displayGame() : void
+ restartLevel() : void
+ checkWin() : boolean
+ changeState(s : String) : void
1
Level
- levelName : String
- obstacles : ArrayList<Obstacle>
- levelLength : int
- progress : double
- completed : boolean
+ loadLevel() : void
+ updateLevel() : void
+ displayLevel() : void
+ addObstacle(o : Obstacle) : void
+ getProgress() : double
contains
Obstacle
- x : double
- y : double
- width : int
- height : int
- speed : double
+ update() : void
+ display() : void
+ moveLeft() : void
+ collide(p : Player) : boolean
1
ui
Button
- x : int
- y : int
- width : int
- height : int
- label : String
+ display() : void
+ isHovered() : boolean
+ isClicked() : boolean
SoundManager
- backgroundMusic : SoundFile
- jumpSound : SoundFile
- deathSound : SoundFile
+ playMusic() : void
+ stopMusic() : void
+ playJump() : void
+ playDeath() : void
spawns
Spike
- damage : int
+ display() : void
+ collide(p : Player) : boolean
Particle
- x : double
- y : double
- velocityX : double
- velocityY : double
- size : double
- lifespan : double
+ update() : void
+ display() : void
+ isDead() : boolean
MovingObstacle
- moveSpeed : double
- direction : int
+ movePattern() : void
+ update() : void
+ display() : void

Meeting 1 Progress Check

Current Functionalities

1. Completed Player Class
   - All methods and variables are complete.
   - Note: Currently has a syntax error preventing the code from running.
2. Completed Obstacle Class
   - All methods and variables are complete.
3. Wrote out skeleton of Spike class
   - Spike is a subclass of Obstacle.
   - I wrote out all of the private fields and method signatures so far.
4. Updated main sketch file (GeometryDash.pde)

Future Functionalities

1. Debug Player class.
   - There is currently an issue with syntax that needs to be fixed.
2. Complete the obstacle subclasses.
   - Finish the methods in Spike class. Create and complete MovingObstacle class. This extends the Obstacle class.
3. Start Game class.
   - If I finish everything else on the list, I will start the Game class, which manages the interactions of the smaller components of the game.

Issues

One issue I had was with initializing the Player constructor. I wasn't sure what values were appropriate for gravity and jumpForce. While coding, I recalled having a similar problem with my Force Simulator lab when I added an additional vector to create a lift force. I went back to the lab and looked at the values I had for the forces there and how they interacted. Doing this was helpful because it gave me an idea for what initial values to put for gravity and the jumpForce. Moving forward, I'll frequently check how the player interacts with other objects, like moving obstacles, so the forces don't overpower each other.

After having created my allObstacles branch, I had a little bit of trouble accessing the branch. However, this was a quick fix because I just had to use git fetch to access the branch through my terminal.

I had difficulties trying to figure out the logic for my collide method in the Player class. I think the most helpful approach was thinking about the pseudocode to break down the steps and then writing the actual code.

Currently, an issue I have on my main branch is that my Player class isn't compiling. This was after I added my collide method. I checked over the syntax for that method and looked over the brackets in the rest of the class. I didn't have a chance to correct this issue before this document submission, because I was mainly focusing on the allObstacles branch. I plan on fixing this issue in the next class period.

Log

I am independently working on the project. The following log is a brief outline of the progress I made, described under Current Functionalities.

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1. Created and completed Player class.
2. Working on main sketch file.
3. Created allObstacles branch.
4. Created and completed Obstacle class.
5. Outlined the Spike class. Currently, working on this class.